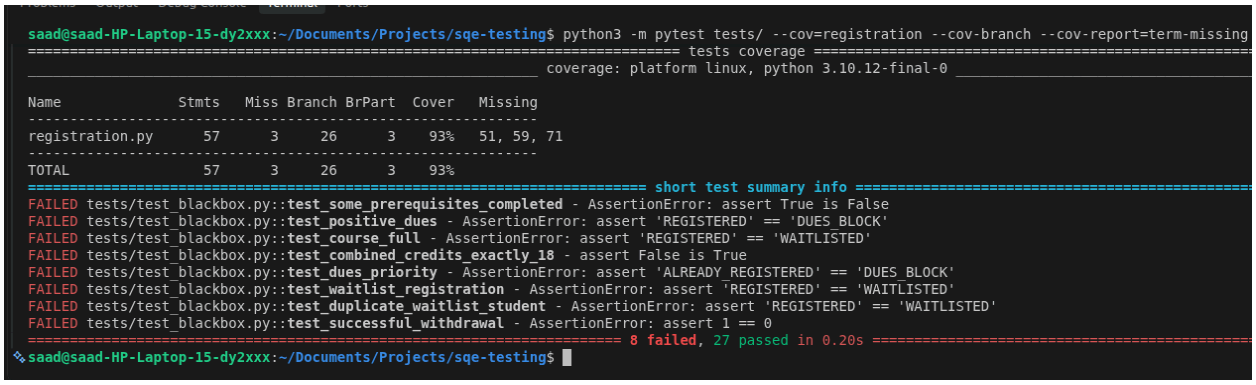
Initial Statement and Branch Coverage From Black-Box Tests



Black-box testing and structural coverage analysis

Initial pytest-cov results show 93% statement coverage and 93% branch coverage. However, there are two separate issues:

1. Some existing black-box tests fail because the implementation does not match the expected requirements.
2. Some execution paths are not covered by the existing tests, requiring structural testing.

It is important not to confuse test failures with coverage gaps. A test can execute a line successfully from a coverage perspective and still fail its assertion.

Identify the gaps and implementation defects

Failed tests reveal these problems:

Test	Problem identified
`test_some_prerequisites_completed`	Implementation accepts any matching prerequisite instead of requiring all.
`test_positive_dues`	Test uses dues = 1, but implementation blocks only dues above 1000.
`test_course_full`	Full course is not treated as waitlisted.
`test_combined_credits_exactly_18`	Implementation uses `< 18`, while the test expects `<= 18`.
`test_dues_priority`	Duplicate registration is checked after dues, but dues of 1 do not trigger the block.
`test_waitlist_registration`	Full-course condition is inconsistent with capacity validation.
`test_duplicate_waitlist_student`	Same unreachable waitlist condition.
`test_successful_withdrawal`	Withdrawal does not decrement `course.enrolled`.

Important structural problem

Consider this code:

if not 0 <= course.enrolled <= course.capacity or course.capacity < 1: raise ValueError("Invalid capacity or enrollment")if course.enrolled > course.capacity: return "WAITLISTED"

The first condition rejects enrollment greater than capacity.

Therefore, the second condition can never be true.

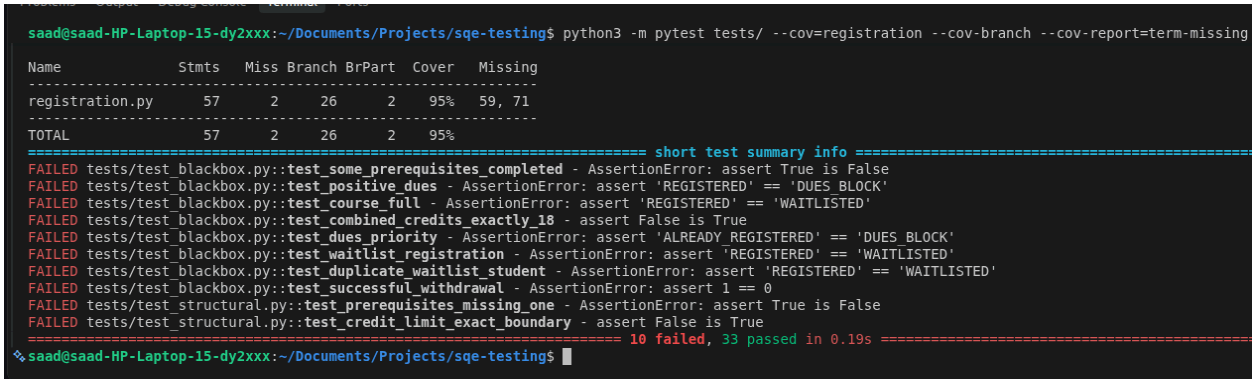
This means the waitlist decision is unreachable through `registration_decision()`.

Similarly, the waitlist insertion inside `register_course()` cannot be reached.

Added structural tests

test_structural.py

After Adding Test Cases



Initially, black-box testing achieved 93% statement and branch coverage. After analyzing the uncovered code paths, additional structural tests were introduced to exercise decision conditions, boundary values, duplicate registration, and withdrawal behavior. The new coverage reached 95%, reducing uncovered statements from 3 to 2 and partially covered branches from 3 to 2. Two source locations, lines 59 and 71, remain uncovered and require further investigation.